

PAPER_924: Black Hole Phonon Ergosphere Superradiance

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Abstract

Extends Kerr black hole superradiance with phonon-vacuum coupling: $\Gamma_{SR} = \Phi_{\{1.25\text{THz}\}} * (m\Omega_H - \omega) \alpha_{BH}$, where Ω_H is the horizon angular velocity, $\alpha_{BH} = (r_g/r)^{2l+2}$ is the BH coupling factor, and Φ is the phonon modulation. The phonon enhancement amplifies the superradiant instability for rotating BHs, creating a new energy extraction channel alongside Penrose process and BZ mechanism. Additionally derives phonon-modified Hawking temperature $T_H^{\text{phonon}} = T_H * (1 + \Phi S_{26}[SSq]/N)$, QPO accretion disk phonon coupling, and phonon-corrected BH entropy.

1. Core Equations

Section A: Lagrangian

$$\Gamma_{SR} = \Phi_{\{1.25\text{THz}\}}(\omega) * (m * \Omega_H - \omega) * \alpha_{BH}$$
$$\Omega_H = a * c / (2 * r_+)$$
$$r_+ = r_g * (1 + \sqrt{1 - a^2})$$
$$\alpha_{BH} = (r_g / r)^{(2\ell + 2)}$$

Section B: VDS/DVP/BH Number Systems

$$T_H^{\text{phonon}} = T_H * (1 + \Phi * S_{26} * [SSq] / N)$$
$$T_H = \hbar * c^3 / (8\pi G M k_B)$$
$$S_{BH}^{\text{phonon}} = S_{BH} * (1 + \text{correction})^2$$

Section SM: SM Anchors

$G = 6.6743e-11 \text{ m}^3/\text{kg}/\text{s}^2$
 $c = 2.998e8 \text{ m/s}$
 $\hbar = 1.055e-34 \text{ J*s}$
 $k_B = 1.381e-23 \text{ J/K}$

2. Parameters

Parameter	Default	Description
M_bh	10 M_sun	BH mass
a_spin	0.9	Kerr spin parameter
m_mode	1	Azimuthal mode number
omega_field	omega_SCm	Field frequency
Phi_0	10^20	Peak phonon amplitude
ell	1	Angular momentum quantum number

3. Key Results

Spin	Gamma_SR	Superradiant?	T_H^{phonon}
$a = 0.1$	< 0	No	$\sim T_H$
$a = 0.9$	> 0	Yes	$\gg T_H$
$a = 0.998$	$\gg 0$	Yes (maximal)	$\gg T_H$

4. Physical Interpretation

Phonon-vacuum coupling amplifies the Kerr BH superradiant instability by providing an additional channel for angular momentum extraction. The condition $m\Omega_H > \omega$ is the standard superradiance threshold; the phonon factor Φ multiplies the gain rate, making superradiance observable at lower spins than GR predicts. The modified Hawking temperature T_H^{phonon} is enhanced by 26-layer phonon coupling, potentially observable in BH analog experiments (sonic horizons in BEC).

5. References

- PAPER_910: Phonon-modulated Hawking temperature (Session 210)
- PAPER_911: QPO accretion disk phonon coupling
- `bh_phonon_interaction.py`: 4-class standalone module